Lesson 11.2 - How to Find Cohen's D to Determine the Effect Size

> **Video:** How to find Cohen's D to determine the Effect Size Using Pooled Standard Deviation by The Organic Chemistry Tutor **Chapter:** 11 - Hypothesis Testing - 2 Samples

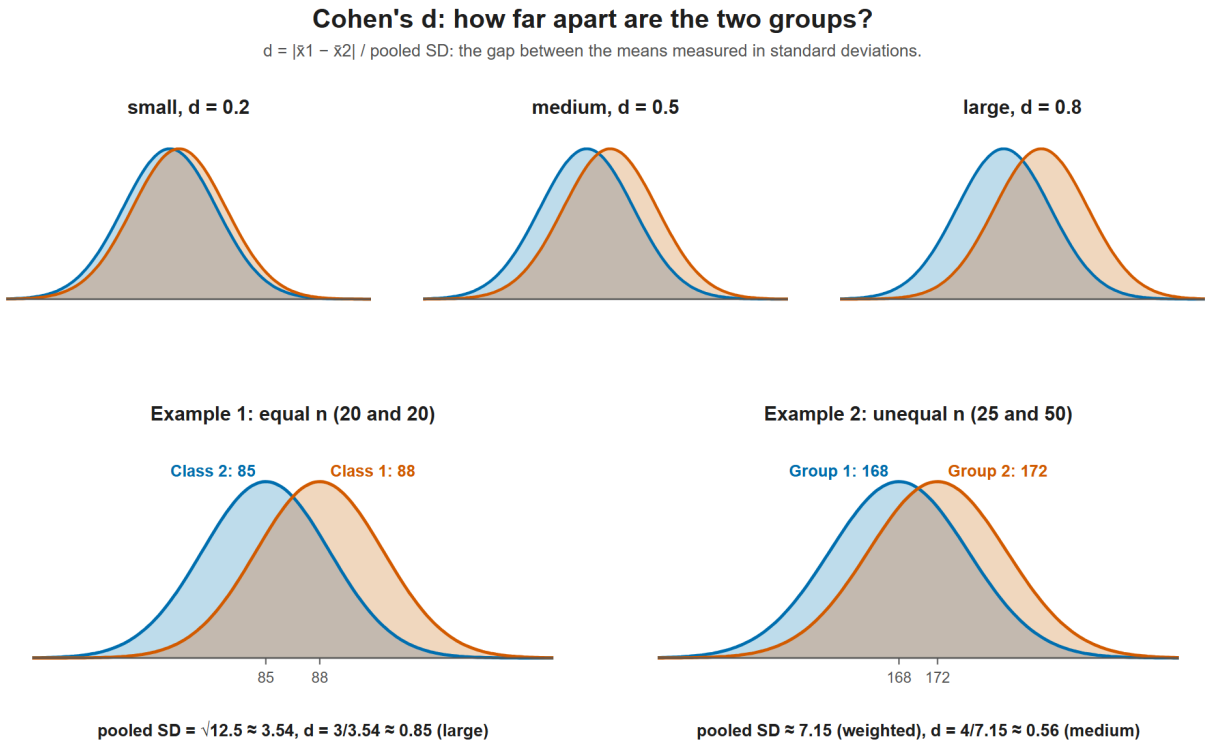
What Cohen's d measures

Cohen's d expresses the difference between two means **in standard-deviation units**:

$$d = \frac{|\bar{x}_1 - \bar{x}_2|}{s_{\text{pooled}}}$$

- $d = 1$: the means are 1 standard deviation apart.
- $d = 2$: 2 standard deviations apart.
- $d = 0.5$: half a standard deviation apart.

d	Effect size
≈ 0.2	Small
≈ 0.5	Medium
≥ 0.8	Large



Cohen's d measures how far apart the two curves sit, in units of the pooled standard deviation. The more the curves overlap, the smaller the effect. Example 1 (d ≈ 0.85) is a large effect; Example 2 (d ≈ 0.56) is medium.

Pooled standard deviation

General formula (use it when the sample sizes differ):

$$s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Shortcut (only when $n_1 = n_2$, or nearly so):

$$s_{\text{pooled}} = \sqrt{\frac{s_1^2 + s_2^2}{2}}$$

Example 1 - two classes, equal sample sizes

	Class 1	Class 2
n	20	20
$\bar{x}$	88	85
s	3	4

Pooled SD (general formula):

$$s_p = \sqrt{\frac{19(3^2) + 19(4^2)}{20 + 20 - 2}} = \sqrt{\frac{171 + 304}{38}} = \sqrt{\frac{475}{38}} = \sqrt{12.5} \approx 3.5355$$

Shortcut (equal n): $\sqrt{\frac{9 + 16}{2}} = \sqrt{12.5} \approx 3.5355$, the same answer ✓

Cohen's d:

$$d = \frac{88 - 85}{3.5355} \approx 0.85 \;\Rightarrow\; \textbf{large effect}$$

Example 2 - two groups, unequal sample sizes

	Group 1	Group 2
n	25	50
$\bar{x}$ (lb)	168	172
s (lb)	5	8

Pooled SD (must use the general formula, because the sizes differ):

$$s_p = \sqrt{\frac{24(5^2) + 49(8^2)}{25 + 50 - 2}} = \sqrt{\frac{600 + 3136}{73}} = \sqrt{\frac{3736}{73}} \approx 7.154$$

The shortcut would give $\sqrt{(25 + 64)/2} \approx 6.671$, which is **wrong** here. Group 2 is twice as large, so its SD (8) should carry more weight, and the correct pooled SD (7.15) is closer to 8.

Cohen's d:

$$d = \frac{|168 - 172|}{7.154} = \frac{4}{7.154} \approx 0.56 \;\Rightarrow\; \textbf{medium effect}$$

Key takeaways

- d = |difference in means| ÷ pooled SD, a **unitless** measure of how big the difference is.
- Benchmarks: 0.2 small, 0.5 medium, 0.8+ large.
- Use the weighted pooled-SD formula unless the sample sizes are equal.
- A hypothesis test asks *whether* there is a difference. Cohen's d asks *how large* it is.